SVAFormer: Integrating Random and Hierarchical Spectral View Attention for Hyperspectral Image Classification

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***Abstract*— Recently, hyperspectral image (HSI) classification methods based on Transformers have developed rapidly. However, these methods still face challenges in handling the widely varying scales and diverse spatial distribution patterns commonly found in HSIs. To address these issues, this article proposes a simple, yet novel HSI classification framework named the spectral view attention Transformer (SVAFormer). Built on the Transformer mechanism, this framework enhances the integration of spectral and spatial features by allowing the spectral token, corresponding to the pixel to be classified, to access spatial neighborhood information from multiple perspectives and levels. Specifically, the framework employs random masking techniques to provide spectral tokens with spatial neighborhood information from different viewpoints, enabling the model to handle diverse land-cover distribution patterns. Additionally, the framework introduces a spectral token-aware pooling layer between adjacent Transformer blocks, which preserves the central role of spectral tokens while progressively expanding the spatial scale represented by each token. This reduces the Transformer's focus on spatially fragmented information and enables spectral tokens to concentrate on spatial neighborhood information at various levels and scales. The key characteristic of this framework is its ability to effectively handle land-cover features of different scales and shapes by strengthening the fusion of spectral and spatial characteristics. Experimental results on multiple public datasets demonstrate that our framework outperforms previous state-of-the-art methods. For the sake of reproducibility, the source code of SVAFormer will be publicly available at https://github.com/chenning0115/SVAFormer.**

***Index Terms*— Attention, deep neural network, hyperspectral image (HSI) classification, Transformer.**

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I. Introduction

HYPERSPECTRAL image (HSI), as an important branch of remote sensing technology, has been widely applied in various fields. By capturing spectral information across hundreds of continuous bands ranging from ultraviolet to infrared, HSIs provide much more detailed spectral data compared to traditional multispectral imaging. This high spectral resolution enables HSIs to demonstrate great potential in areas such as vegetation monitoring [1], [2], geological mapping [3], [4], urban planning [5], [6], and soil characterization [7], [8]. For instance, in vegetation monitoring, HSIs can accurately distinguish different types of vegetation and monitor their health and coverage changes. In geological mapping, it is used to identify and classify surface rocks and mineral types, providing detailed data for geological structure analysis and mineral resource exploration. In urban planning, HSIs help analyze urban heat island effects, environmental pollution, and surface material distribution, assisting in the planning and management of urban environments. These extensive applications not only enhance the efficiency of various industries but also promote the advancement of scientific research.

In HSI classification, traditional classification techniques played a significant role in the early handling of high-dimensional spectral data. Pal and Foody [9] and Melgani and Bruzzone [10] argued that HSIs, with their small training samples and excessive features, often experienced reduced classification accuracy; thus, feature extraction was necessary before training with support vector machines (SVMs) to improve classification performance. Huang et al. [11] suggested that K-nearest neighbors (KNNs) had significant advantages in processing hyperspectral data, enabling efficient and accurate classification results. Velásquez et al. [12] found that decision trees were advantageous for handling hyperspectral data with similar appearance contours, providing fast and reliable classification results. Bandos et al. [13] believed that combining linear discriminant analysis with regularization yielded good classification results for small training samples. However, while these traditional HSI classification methods offered effective tools for data analysis and processing, they faced significant challenges in addressing the complexity and high dimensionality of HSIs, such as rapidly increasing computational complexity, overfitting, and the curse of dimensionality.

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Compared to traditional HSI classification methods, convolutional neural networks (CNNs) are better at integrating spectral and spatial features and handling high-dimensional data, thereby improving classification accuracy and robustness [14]. Yue et al. [15] suggested that CNNs can automatically extract complex features, efficiently utilize spatial information, and combine it with spectral features to enhance classification precision. Zhao and Du [16] and Luo et al. [17] proposed a deep CNN framework that integrates multilevel features, significantly improving HSI classification accuracy. Lee and Kwon [18] further introduced a CNN-based HSI classification method that can be trained with limited samples. Bhatti et al. [19] proposed a method to extract deep spectral features by using local similarity projection (LSP) and filtering for dimensionality reduction, followed by CNNs to extract effective features, thereby improving classification accuracy. Guo et al. [20] also emphasized the importance of dimensionality reduction, suggesting the use of Factor analysis and wavelet transformation for hyperspectral data, followed by CNNs for extracting spatial–spectral fusion features. Vaddi and Manoharan [21] utilized probabilistic principal component analysis (PPCA) for spectral feature extraction and Gabor filters for spatial feature extraction and then used CNNs for spatial–spectral fusion feature extraction. Many of these methods rely on either single 2-D-CNNs or 3-D-CNNs, while Yu et al. [22] and Ge et al. [23] combined both to leverage their respective advantages, effectively improving HSI classification accuracy. Ding et al. [24] combined graph convolutional networks (GCNs) with CNNs to utilize their strengths for extracting spatial–spectral fusion features and enhancing HSI classification accuracy. However, these methods often require complex network architectures, struggle with noise issues, and exhibit limited transferability. These limitations affect the model's generalization ability across different datasets or scenarios.

The Transformer is a deep learning model based on self-attention mechanisms, differing from traditional CNNs that rely on local receptive fields. By processing sequence data in parallel, Transformers can model complex interactions between distant regions in an image, thereby enhancing classification performance [25]. Following the tremendous success of Transformers in semantic recognition, their superior feature representation capabilities and efficient parallel processing have attracted widespread attention [26]. Dosovitskiy et al. [27] introduced the Vision Transformer (ViT), demonstrating that Transformers also perform exceptionally well in natural image classification tasks. Moreover, Transformers have shown their superiority in medical image classification and nonhyperspectral radar image classification. In medical image classification, Transformers achieve precise classification of complex medical images through their powerful feature extraction and global context understanding abilities [28], [29]. Unlike optical images, radar images typically contain various types of noise and interference and capture information in the form of radar echoes, which are influenced by material properties and environmental conditions. Transformers, with their sequence data processing and attention mechanism capabilities, can adapt to these unique features. For nonhyperspectral radar images, Transformers help extract and integrate features from different radar bands or time slices, thereby improving classification robustness and accuracy under various complex conditions [30], [31].

However, HSIs differ significantly from natural images and nonhyperspectral radar images. Natural images consist of three color channels (RGB), and nonhyperspectral radar images typically have only one spectral channel, capturing limited spectral information. In contrast, HSIs contain rich spectral information across hundreds of contiguous bands, with each band providing detailed data about the material composition and characteristics of the observed scene. Applying Transformers to HSI classification must address the high-dimensional nature of the data, where each pixel is represented by a multidimensional spectral vector [32]. The challenge lies in effectively modeling both the spatial and spectral dimensions of hyperspectral data. Transformers, with their ability to handle high-dimensional inputs and capture complex spectral interactions, offer promising solutions for improving classification accuracy. They leverage self-attention mechanisms to focus on relevant spectral features while integrating spatial context, which is crucial for distinguishing materials with subtle spectral differences. This necessitates different approaches when applying Transformers to HSI classification. In [33], a Transformer-based HSI classification framework was proposed, which effectively learns spectral local sequence information and introduces cross-layer skip connections to reduce information loss. Works in [34] and [35] utilized CNNs to extract surface features and then leveraged Transformers to further extract deep features, thereby improving classification accuracy. In [36], CNNs and Transformers were used to extract spatial and spectral features, respectively, taking advantage of each approach to enhance the accuracy of HSI classification. Inspired by ViT, the study in [37] proposed a Transformer-based method capable of effectively learning spatial–spectral features for HSI classification. In [38], a wavelet-based CNN feature extractor was introduced into the ViT framework, enabling the Transformer to fully utilize the spatial information in HSIs, such as texture and shape details. This integration enhances the model's ability to capture fine-grained spatial features, which is crucial for accurate HSI classification. In [39], a novel Transformer-based HSI classification method centered around the central pixel was creatively proposed. This method uses random rotation and position encoding based on the central pixel, significantly improving classification accuracy.

However, existing methods still face challenges in addressing the common issues of variable scales and diverse spatial distribution patterns in HSIs. This limitation reduces the model's generalization ability when handling spatial position perturbations. To tackle this problem, we propose a novel HSI classification framework called spectral view attention Transformer (SVAFormer). This framework consists of two modules: the **random spectral view attention** (RSVA) block and the **hierarchical spectral view attention** (HSVA) block. The RSVA module introduces a random mask that leverages the spectral token, infusing it with spatial neighborhood information from various perspectives. This enables the model to

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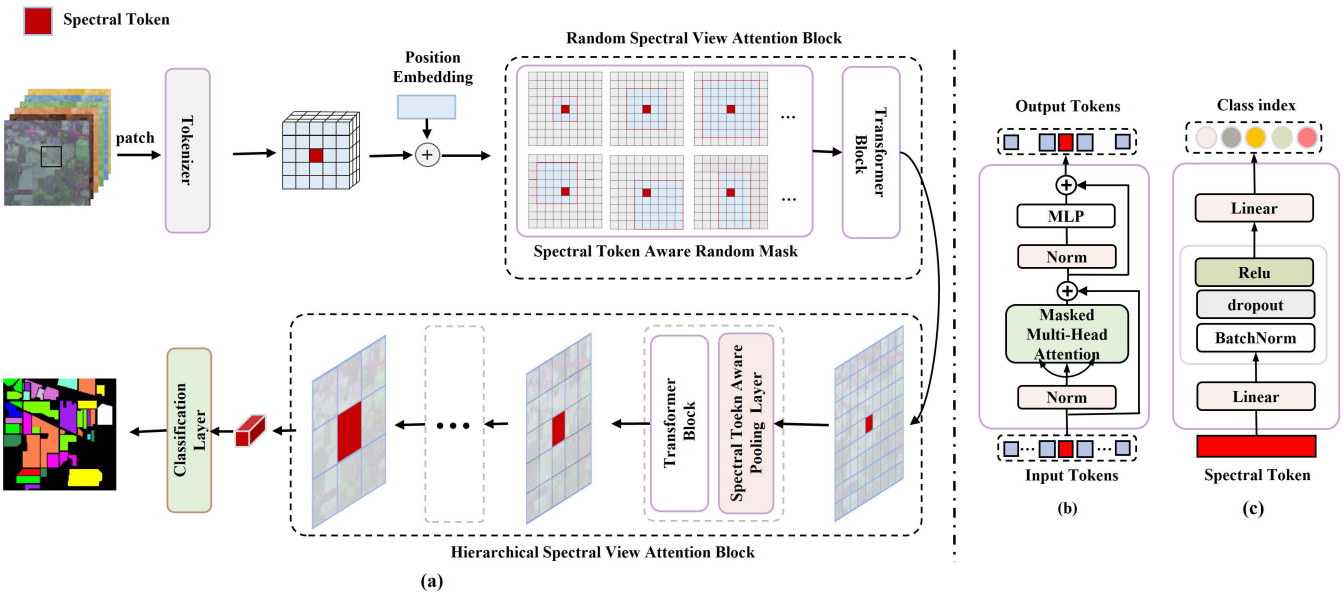


Fig. 1. Framework of SVAFormer. (a) Overall structure of the proposed model. (b) Internal structure of the Transformer block. (c) Structure of the classification layer.

effectively manage diverse land-cover distribution patterns. Meanwhile, the HSVA module addresses the multiscale problem of land covers by utilizing multilevel pooling. Together, these two modules allow the SVAFormer framework to focus on relevant information across various shapes and scales, thereby improving classification accuracy and enhancing the model's generalization ability. The specific innovations are as follows.

1) This article proposes the RSVA block, a module that, based on spectral tokens, employs random masking techniques to provide spatial neighborhood information from different perspectives. This simple method enhances the Transformer's ability to focus on local spatial information, enabling the model to handle diverse land-cover distribution patterns. Additionally, this method enriches the perspective information available to the limited hyperspectral training samples, thereby effectively enhancing the model's generalization capabilities.
2) This article proposes the HSVA block, a module that introduces a spectral token-aware pooling layer between adjacent Transformer blocks. This allows the model to progressively expand the spatial scale represented by each token while preserving the central role of the spectral tokens. Consequently, the model reduces its attention to spatially fragmented information, enabling spectral tokens to focus on spatial neighborhood information at different levels and scales. Ultimately, this approach effectively improves classification accuracy.
3) This article proposes a novel HSI classification framework named SVAFormer. Unlike previous methods, this Transformer-based framework allows the spectral token of a pixel to access spatial neighborhood information from multiple perspectives and levels. This enhances spatial feature richness and improves the integration of spectral and spatial features, ultimately boosting classification accuracy. Experimental evaluations on several public datasets demonstrate that our framework outperforms previous state-of-the-art algorithms in classification performance.

II. Methodology

A. Overview

The overall structure of the proposed SVAFormer is illustrated in Fig. 1. Given an HSI denoted as $\mathcal{H} \in \mathbb{R}^{B\times H\times W}$, where H and W represent the height and width, respectively, and B represents the number of spectral channels. During the data preprocessing stage, we first perform dimensionality reduction on the raw input data using principal component analysis (PCA), resulting in the processed data $X \in \mathbb{R}^{C\times H\times W}$, where C is the dimensionality of the reduced channels. Generally, in HSI classification tasks, to effectively incorporate spectral–spatial information, we typically select a small patch centered on the pixel to be classified as the model input, denoted as $x \in \mathbb{R}^{C\times w\times w}$, where w represents the patch size. Assuming we have N training sample patches, $X_N = \{x_1, x_2, x_3, \ldots, x_N\}$, with corresponding central pixel class labels $Y_N = \{y_1, y_2, y_3, \ldots, y_N\}$.

After data preprocessing, we first tokenize the input patch X_N to adapt it for processing by Transformer-type networks. Specifically, we split the HSI along the spatial pixel dimensions and used a CNN to convert the pixel information into token vectors. Notably, since the central pixel of the patch is the target pixel for classification and contains crucial spectral information in HSIs, we designate the token corresponding to this pixel as the **spectral token**. Furthermore, to accurately encode the positional information of different pixels, we apply position embedding to the pixels, and the encoded vector Z is used as the input for the subsequent network.

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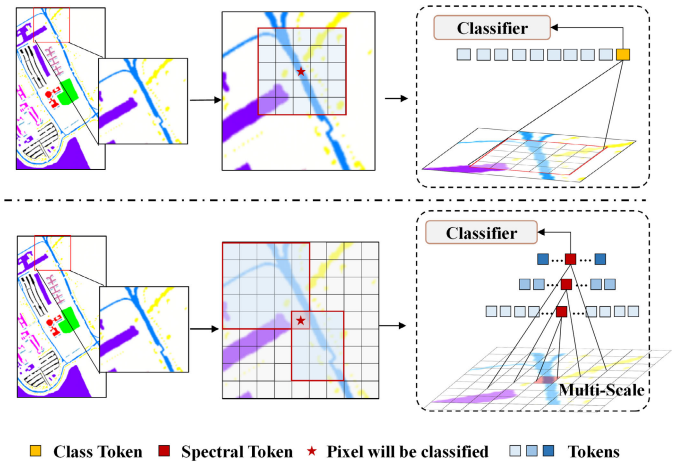


Fig. 2. Differences in workflow between traditional ViT-based HSI classification methods and SVAFormer. The top diagram illustrates the workflow of ViT-based algorithms, while the bottom diagram shows the workflow of SVAFormer. By leveraging spectral tokens with diverse perspectives and multilevel attention, SVAFormer effectively adapts to objects with varying distribution patterns.

To effectively integrate the spectral information contained in the spectral token with the spatial information contained in the tokens corresponding to neighboring pixels, we sequentially input Z into the RSVA and HSVA. The RSVA, considering the spectral token, uses a random masking technique to provide the spectral token with spatial neighborhood information from different perspectives, thereby enhancing the Transformer's ability to focus on local spatial information. The HSVA introduces a spectral token-aware pooling layer between adjacent Transformer blocks, allowing the model to maintain the prominence of the spectral token while progressively enlarging the spatial scale represented by a single token. This reduces the Transformer's attention to spatially fragmented information and enables the spectral token to capture spatial neighborhood information at different levels and scales.

Finally, we extract the spectral token representing the central pixel from the output vector after the RSVA and HSVA modules. This token is then fed into a classification layer to classify the pixel and obtain the probability P of it belonging to different land-cover classes. Consistent with traditional classification algorithms, we use the cross-entropy (CE) loss as the final classification loss, as shown in (1), while $p_{ij} \in P$, with K being the class num.

Fig. 2 illustrates the differences in workflow between traditional ViT-based HSI classification methods and SVAFormer. Specifically, SVAFormer leverages RSVA and HSVA to provide spectral tokens with different perspectives and multilevel attention, thereby enabling it to easily adapt to land covers with varying distribution patterns

$$\text{CEloss} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{j=1}^{K}\left(y_{ij}\log\left(p_{ij}\right)\right). \tag{1}$$

B. Random Spectral View Attention Mechanism

This article proposes the RSVA block, which, considering the spectral token, uses random masking techniques to provide the spectral token with spatial neighborhood information from different perspectives. As previously mentioned, let the input of the RSVA block be $Z \in \mathbb{R}^{C\times w\times w}$ and the output be $Z^{(O1)}$. In $Z^{(O1)}$, the unmasked part is referred to as the reserve patch, denoted as Z^R, while the masked part is denoted as Z^M. As illustrated in Fig. 3, the RSVA includes two types of random masking methods: one keeps the spectral token at the center of Z^R after masking, while the other places the spectral token at a random position within Z^R after masking. The former method enhances the ability of the spectral token to attend to different spatial ranges during modeling, while the latter mainly strengthens the spectral token's ability to extract spatial information under varying spatial scales and orientations, reducing the path dependency caused by always having the target pixel at the center of the patch and enhancing the model's classification capability under different spatial distribution scenarios. In practice, during the training process, we randomly select which masking scheme to use with a probability threshold of ω. During serving, we opt for the former masking method and fix the serving patch size to maintain stability. The serving patch refers to the fixed central area of Z^R, which ensures that the model observes a consistent spatial range around the spectral token during inference. This helps prevent fluctuations in the model's output by maintaining a stable region for attention computation. It is important to note that, in both masking methods, the spectral token is always present in Z^R. RSVA effectively enhances the Transformer's ability to attend to local spatial information. By using a random masking technique, it provides diverse perspective information to the limited hyperspectral training samples, enabling the model to handle various land-cover patterns. This, in turn, improves the model's adaptability and generalization capability for different types of ground objects.

C. Hierarchical Spectral View Attention Mechanism

This article proposes the HSVA block, which introduces a spectral token-aware pooling layer between adjacent Transformer blocks. This allows the model to maintain the prominence of the spectral token while progressively enlarging the spatial scale represented by a single token, thereby reducing the Transformer's attention to spatially fragmented information. Consequently, the spectral token can focus on spatial neighborhood information at different levels and scales. Specifically, for the input $Z^{(O1)}$, to ensure that the central position of the spectral token remains unchanged, we constrain common pooling operators as shown in (2) and (3). Fig. 4 illustrates the differences between the HSVA and the ViT in processing hyperspectral remote sensing images. Unlike ViT, where the number of pixel tokens remains stable across different Transformer layers, the HSVA blocks progressively reduce the number of pixels, enlarging the spatial scope represented by each pixel. Moreover, the HSVA preserves the central position of the pixel represented by the spectral token, ensuring that it remains unchanged and ultimately serves as the input embedding for the subsequent classification layer

$$Z^{O2} = \text{pooling}(\text{kernal} = 3, \text{stide} = 2 \\ \text{padding} = \text{pad})\left(Z^{O1}\right) \tag{2}$$

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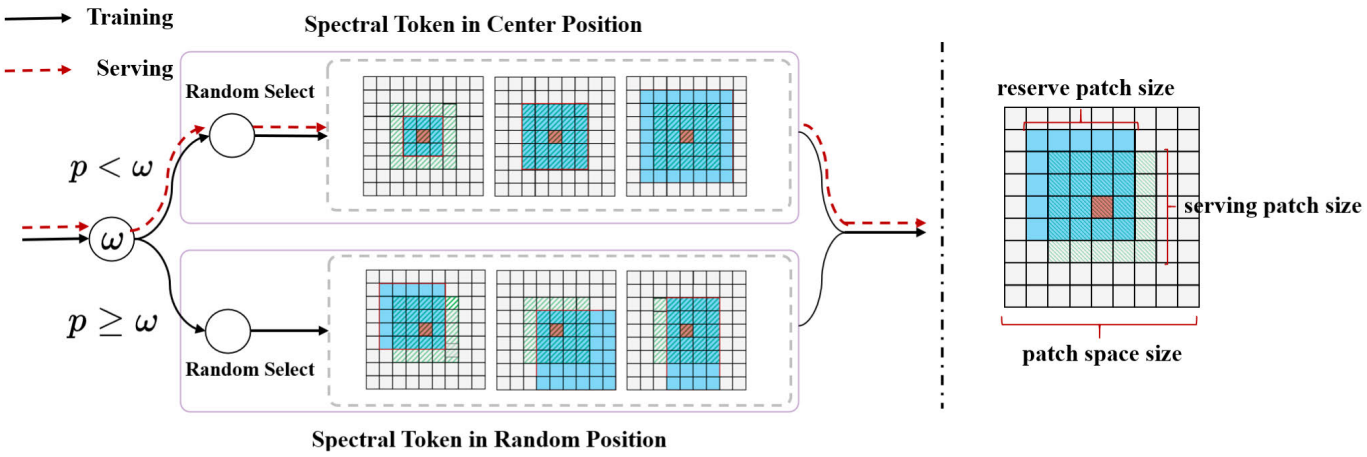


Fig. 3. Details of the RSVA block. The light gray area represents the input patch size, referred to as the patch space size. The blue area represents the nonmasked region retained after applying random masks during model training. The green striped area indicates the nonmasked region retained during serving, showing the actual serving patch size. During the training process, we randomly select the masking scheme to use based on a probability threshold ω. During serving, we use the former masking method and fix the serving patch size to maintain stability.

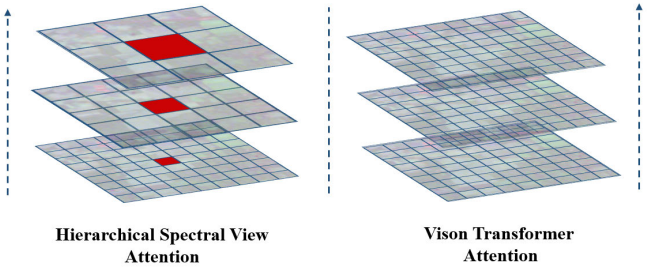


Fig. 4. Comparison of HSVA and ViT for HSI classification process.

$$\text{pad} = \begin{cases} 0 & (h-1)\%2 \neq 0 \\ 1 & (h+1)\%2 \neq 0 \end{cases}. \tag{3}$$

III. Experiments

A. Experimental Settings

1) Datasets: To verify the effectiveness of our algorithm, we applied the algorithm to three public datasets: the Indian Pines (IP) dataset, the Pavia University (PU) dataset, and the WHU-Hi-HongHu (WHHH) dataset.

The IP dataset, collected in 1992 using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor, covers the northwestern region of Indiana, United States. This dataset comprises 224 spectral bands in the wavelength range of 0.4–2.5 μm, consisting of 145 $\times$ 145 pixels, each with a spatial resolution of 20 m. It includes 16 distinct land-cover classes. We excluded 24 bands associated with water absorption and noise, selecting a subset of 200 bands. The false color image and the classification ground truth are shown in Fig. 5.

The PU dataset, acquired in 2001 using the Reflective Optics System Imaging Spectrometer (ROSIS) sensor, covers the geographical area of PU in Northern Italy. It consists of 115 spectral bands, with 12 noise bands excluded, resulting in 103 bands used for analysis. The dataset spans a spatial domain of 610 $\times$ 340 pixels, with a spatial resolution of 1.3 m,

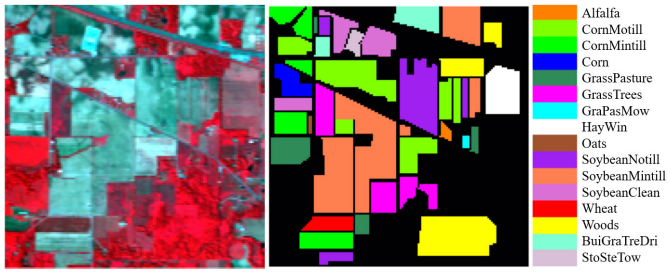


Fig. 5. False-color composite image, ground-truth map, and legend of the IP dataset.

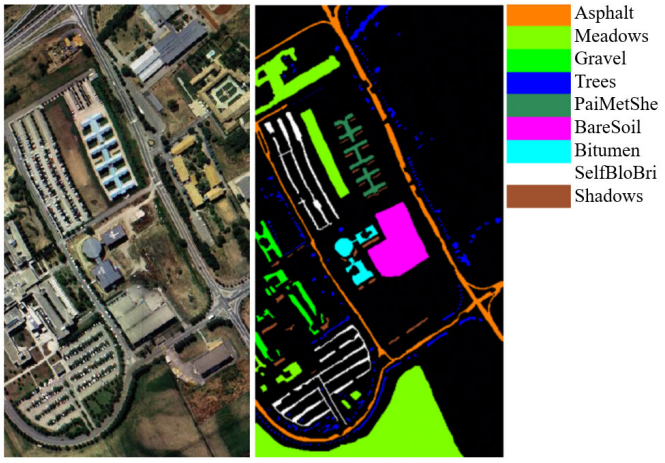


Fig. 6. False-color composite image, ground-truth map, and legend of the PU dataset.

and includes nine distinct land-cover classes. The false color image and the classification ground truth are shown in Fig. 6.

The WHHH dataset [40], [41] was collected on November 20, 2017, from 16:23 to 17:37 in Honghu City, Hubei Province, China, using a Headwall Nano-Hyperspec imaging sensor with a 17-mm focal length. The sensor was mounted on a DJI Matrice 600 Pro unmanned aerial vehicle (UAV) platform, which operated at an altitude of 100 m, capturing images with a resolution of 940 $\times$ 475 pixels. The dataset

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TABLE I
TRAINING AND TEST SAMPLE NUMBERS IN THE IP DATASET, THE PU DATASET, AND THE WHHH DATASET

No.	Indian Pines			Pavia University			WHU-Hi-HongHu dataset		
	Class	Train.	Test.	Class	Train.	Test.	Class	Train.	Test.
1	Alfalfa	10	36	Asphalt	10	6621	Red roof	10	14031
2	CornMotill	10	1418	Meadows	10	18639	Road	10	3502
3	CornMintill	10	820	Gravel	10	2089	Bare soil	10	21811
4	Corn	10	227	Trees	10	3054	Cotton	10	163275
5	GrassPasture	10	473	PaintedMetalSheets	10	1335	Cotton firewood	10	6208
6	GrassTrees	10	720	BareSoil	10	5019	Rape	10	44547
7	GrassPastureMowed	10	18	Bitumen	10	1320	Chinese cabbage	10	24093
8	HayWindrowed	10	468	SelfBlockingBricks	10	3672	Pakchoi	10	4044
9	Oats	10	10	Shadows	10	937	Cabbage	10	10809
10	SoybeanNotill	10	962				Tuber mustard	10	12384
11	SoybeanMintill	10	2445				Brassica parachinensis	10	11005
12	SoybeanClean	10	583				Brassica chinensis	10	8944
13	Wheat	10	195				Small Brassica chinensis	10	22497
14	Woods	10	1255				Lactuca sativa	10	7346
15	BuildingsGrassTreesDrives	10	376				Celtuce	10	992
16	StoneSteelTowers	10	83				Film covered lettuce	10	7252
17							Romaine lettuce	10	3000
18							Carrot	10	3207
19							White radish	10	8702
20							Garlic sprout	10	3476
21							Broad bean	10	1318
22							Tree	10	4030
	Total	160	10089	Total	90	42686	Total	220	386473

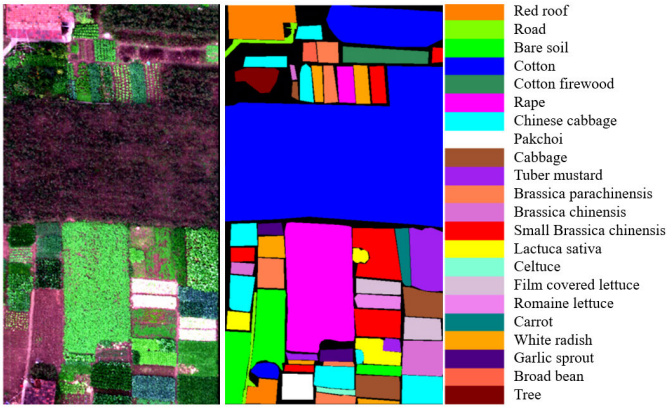


Fig. 7. False-color composite image, ground-truth map, and legend of the WHHH dataset.

includes 270 spectral bands ranging from 400 to 1000 nm, resulting in a spatial resolution of approximately 0.043 m for the UAV-borne hyperspectral imagery. The images depict a complex agricultural scene with various crop types, including different varieties of the same crop, such as Chinese cabbage, cabbage, Brassica chinensis, and small Brassica chinensis. Overall, the dataset encompasses 22 distinct land-cover classes. The false color image and the classification ground truth are shown in Fig. 7.

Table I shows the size of the training and testing datasets used in this experiment, including the sample distribution of each land-cover category.

2) Evaluation Metrics: We will compare the effectiveness of our algorithm with other algorithms from four aspects, mainly including overall accuracy (OA), average accuracy (AA), kappa coefficient (κ), and the classification accuracy of each land-cover category itself. Specifically, OA represents the proportion of correctly classified samples across all classes. It provides a general measure of the classification performance; AA is the average of the individual class accuracies, which accounts for imbalanced class distributions. It indicates how well each class is classified, on average. The Kappa coefficient measures the agreement between the predicted and true classifications, adjusting for the chance agreement. It is particularly useful for evaluating classification performance when there is a class imbalance. These metrics are widely used in HSI classification because they provide a comprehensive evaluation of model performance, considering both OA and class-specific details.

3) Training Details: The PyTorch training framework was utilized for the implementation and training of the model, with the basic hardware environment comprising an Intel[1] Xeon[1] Platinum 8336C CPU @ 2.30 GHz, 64 cores, 256 GB of memory, and two NVIDIA GeForce RTX 4090 GPUs, each with 24 GB of memory.

4) Parameter Analysis: We analyzed key parameters, including the (patch space size, serving patch size) pair, ω, and the depth of the Transformer. Figs. 8 and 9 present the experimental results for finding the optimal parameter settings, revealing that different datasets have different optimal parameter choices. For various (patch space size, serving patch size) combinations, the optimal parameters for the IP, PU, and WHHH datasets are (17, 9), (25, 25), and (25, 25), respectively. Additionally, for the PU and WHHH datasets, we observed that as both the patch space size and serving

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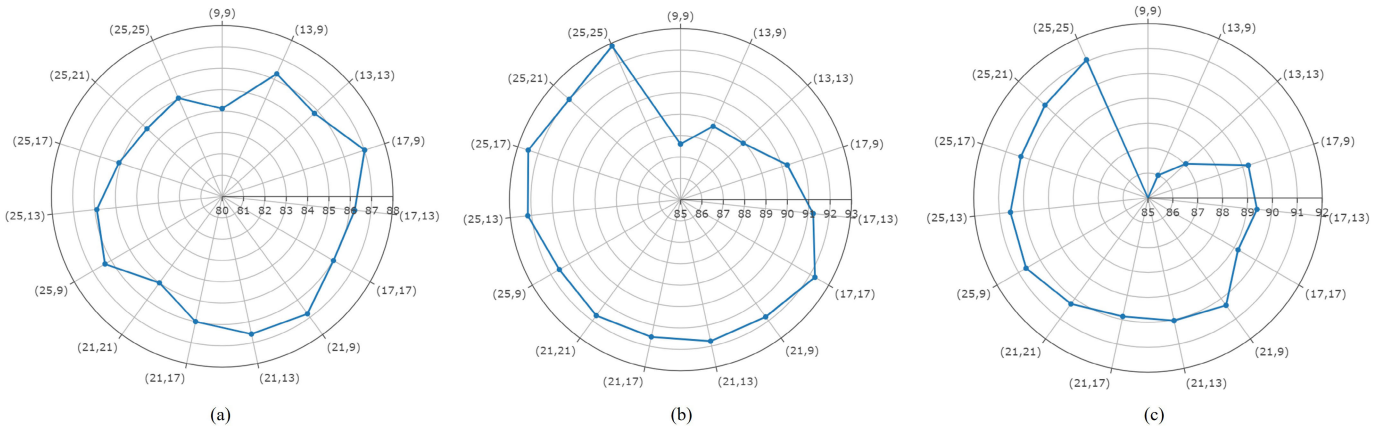


Fig. 8. Classification accuracy on IP, PU, and WHHH datasets for various patch space and serving patch sizes. (a) Results on the IP dataset. (b) Results on the PU dataset. (c) Results on the WHHH dataset. The radial direction represents OA (%), while the rotational direction corresponds to combinations of patch space size and serving patch size (e.g., patch space size and serving patch size).

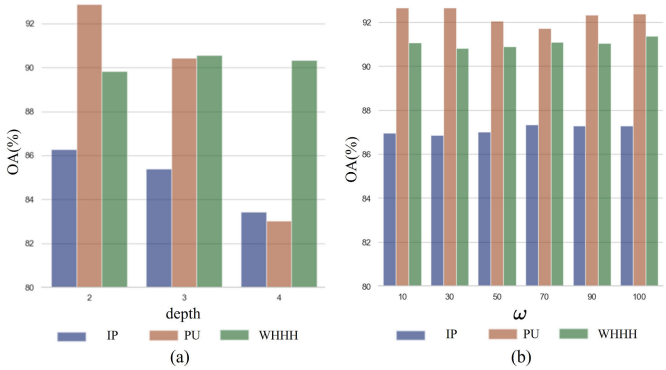


Fig. 9. Classification accuracy on the IP, PU, and WHHH datasets under different depths and ω. (a) Results under varying depths. (b) Results under varying ω.

patch size increase, the corresponding OA also increases. This indicates that larger patch sizes better showcase the advantages of our algorithm, effectively handling information extraction over larger spatial regions. The reason our method performs well with larger patch sizes is that the RSVA and HSVA modules effectively mitigate the typical challenges faced by Transformer-like models when dealing with larger input patches. As the patch size increases, traditional Transformer methods experience a rapid increase in token sequence length, which can reduce the effectiveness of the attention mechanism and degrade performance. However, our approach leverages the RSVA module's random masking technique to reduce the effective token sequence length while preserving crucial spectral information. Additionally, the HSVA module continuously reduces the number of tokens passed between layers using the spectral token-aware pooling layer. This design allows our method to maintain attention efficiency and expand the spatial information acquisition range, which results in improved performance as the patch size increases. Conversely, this trend is not apparent for the IP dataset, which we believe may be due to its inherently lower spatial resolution.

Regarding the ω parameter, the optimal values for the IP, PU, and WHHH datasets are 70, 10, and 100, respectively. For the PU dataset, we found that a mask scheme without a fixed spectral token center position is more effective. We attribute this to the presence of elongated features such as Asphalt in the PU dataset, which cannot be effectively represented by a square patch centered on the spectral token.

The depth of the Transformer block also impacts the three datasets differently. For the IP and PU datasets, the OA tends to decrease with increasing depth, while the OA for the WHHH dataset remains relatively stable. The optimal depths for the IP, PU, and WHHH datasets are two, two, and three layers, respectively.

B. Performance Analysis

1) SpectralFormer [33] is a widely used Transformer-based algorithm and one of the most frequently used benchmarks in recent years.
2) Hyperspectral image Transformer (HiT) [35], based on the ViT approach, incorporates convolution operations within the Transformer to process spectral–spatial information, making it a notable Transformer-based algorithm for HSI classification tasks.
3) Spatial-spectral 1D swin transformer (SS1DSwin) [42] features a group-wise feature tokenization module (GFTM) and a 1-D Swin Transformer with a cross-block normalized connection module (TCNCM). It is one of the latest notable Transformer-based algorithms for HSI classification.
4) Spatial-spectral feature tokenization transformer (SSFTT) [34] combines the CNN and Transformer techniques and introduces a Gaussian-weighted feature tokenizer to further enhance performance. It is one of the Transformer-based algorithms with competitive effectiveness in current HSI classification tasks.
5) Spatial–spectral transformer with cross-attention (CASST) [43] integrates spectral–spatial information through the Cross-Attention module and is one of the Transformer-based algorithms with competitive effectiveness in current HSI classification tasks.
6) Light self-Gaussian-attention (LSGA) [44] is one of the latest Transformer-based algorithms that is a ViT based on the LSGA mechanism, which extracts global deep

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TABLE II
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE IP DATASET (TEN LABELS PER CLASS)

Class	SpectralFormer(2022)	Hit(2022)	SS1DSwin(2023)	SSFTT(2022)	CASST(2022)	LSGA(2023)	MASSFormer(2024)	Ours
1	87.22 ± 11.21	99.44 ± 1.24	88.89 ± 5.20	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	98.67 ± 1.42
2	24.30 ± 4.44	40.42 ± 11.73	38.60 ± 4.32	75.98 ± 2.67	80.37 ± 2.58	49.37 ± 3.59	65.53 ± 5.25	$\mathbf{83.08 \pm 6.83}$
3	52.07 ± 5.52	31.17 ± 6.67	62.44 ± 12.55	84.63 ± 1.58	80.15 ± 3.19	84.17 ± 1.81	$\mathbf{87.05 \pm 3.76}$	81.03 ± 8.40
4	55.77 ± 13.32	63.35 ± 24.62	61.06 ± 11.50	93.30 ± 3.83	89.07 ± 2.30	$\mathbf{99.91 \pm 0.20}$	97.89 ± 2.52	98.71 ± 1.31
5	50.15 ± 15.23	38.09 ± 4.69	79.20 ± 10.62	$\mathbf{94.71 \pm 1.11}$	92.56 ± 1.13	84.14 ± 1.90	82.28 ± 10.40	82.53 ± 8.47
6	54.92 ± 3.79	73.22 ± 9.47	63.33 ± 29.82	93.36 ± 0.58	96.50 ± 1.96	95.83 ± 1.47	95.58 ± 1.41	$\mathbf{98.88 \pm 0.78}$
7	87.78 ± 15.42	$\mathbf{100.00 \pm 0.00}$	88.89 ± 8.78	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$
8	82.39 ± 8.86	98.58 ± 1.31	92.69 ± 2.03	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	99.91 ± 0.19	99.62 ± 0.75	99.96 ± 0.21
9	$\mathbf{100.00 \pm 0.00}$	98.00 ± 4.47	96.00 ± 8.94	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$
10	52.93 ± 9.01	60.09 ± 16.08	56.11 ± 4.21	79.83 ± 3.65	75.24 ± 1.48	79.50 ± 2.66	72.64 ± 12.79	$\mathbf{87.11 \pm 8.55}$
11	32.25 ± 10.66	56.92 ± 9.22	44.79 ± 7.99	67.46 ± 4.00	65.43 ± 1.36	69.22 ± 3.73	69.90 ± 3.18	$\mathbf{77.39 \pm 8.34}$
12	26.52 ± 6.36	32.71 ± 10.14	51.70 ± 7.91	81.37 ± 6.01	81.17 ± 5.04	81.06 ± 5.54	74.99 ± 5.30	$\mathbf{89.80 \pm 3.83}$
13	74.36 ± 21.55	81.54 ± 5.54	97.13 ± 0.58	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	$\mathbf{100.00 \pm 0.00}$	99.18 ± 1.83	99.77 ± 0.54
14	62.71 ± 22.16	84.01 ± 16.40	83.38 ± 5.89	$\mathbf{97.91 \pm 0.51}$	95.38 ± 2.29	89.96 ± 2.24	95.62 ± 4.48	96.16 ± 4.32
15	38.56 ± 10.61	44.30 ± 8.24	78.14 ± 12.30	98.35 ± 0.55	96.86 ± 1.57	85.21 ± 4.42	95.80 ± 3.18	$\mathbf{99.12 \pm 1.34}$
16	97.59 ± 2.95	93.49 ± 4.48	98.55 ± 1.98	$\mathbf{100.00 \pm 0.00}$	99.28 ± 1.62	$\mathbf{100.00 \pm 0.00}$	99.76 ± 0.54	$\mathbf{100.00 \pm 0.00}$
OA(%)	45.44 ± 3.35	58.79 ± 4.74	60.14 ± 5.04	83.30 ± 1.35	82.26 ± 0.59	78.24 ± 0.92	80.82 ± 1.29	$\mathbf{87.33 \pm 2.30}$
AA(%)	61.22 ± 1.85	68.46 ± 3.02	73.81 ± 3.36	91.68 ± 0.63	90.75 ± 0.57	88.64 ± 0.60	89.74 ± 0.62	$\mathbf{93.26 \pm 0.76}$
$\kappa \times 100$	39.90 ± 3.41	53.24 ± 5.16	55.64 ± 5.48	81.20 ± 1.49	80.04 ± 0.65	75.41 ± 1.00	78.34 ± 1.43	$\mathbf{85.70 \pm 2.54}$

TABLE III
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE PU DATASET (TEN LABELS PER CLASS)

Class	SpectralFormer(2022)	Hit(2022)	SS1DSwin(2023)	SSFTT(2022)	CASST(2022)	LSGA(2023)	MASSFormer(2024)	Ours
1	38.18 ± 5.61	58.06 ± 7.64	64.36 ± 8.47	71.89 ± 4.95	85.51 ± 2.00	81.52 ± 0.86	82.34 ± 6.66	$\mathbf{85.90 \pm 5.53}$
2	75.18 ± 1.96	85.53 ± 6.78	72.44 ± 3.73	89.71 ± 1.87	91.91 ± 1.15	$\mathbf{94.15 \pm 1.81}$	86.29 ± 3.89	91.95 ± 7.12
3	60.58 ± 1.33	42.72 ± 15.02	57.78 ± 11.16	90.33 ± 1.49	92.99 ± 4.33	88.84 ± 3.47	91.51 ± 4.44	$\mathbf{95.55 \pm 4.63}$
4	87.25 ± 4.29	92.52 ± 3.06	85.80 ± 1.90	93.46 ± 2.60	83.24 ± 4.78	92.55 ± 2.41	$\mathbf{96.01 \pm 1.49}$	94.71 ± 1.81
5	99.94 ± 0.03	99.97 ± 0.04	99.99 ± 0.03	99.19 ± 1.28	98.58 ± 1.17	$\mathbf{100.00 \pm 0.00}$	99.94 ± 0.13	99.99 ± 0.04
6	54.18 ± 7.07	60.62 ± 14.84	63.34 ± 8.30	97.48 ± 1.85	$\mathbf{97.68 \pm 2.19}$	96.17 ± 2.70	90.53 ± 7.46	96.33 ± 3.60
7	91.61 ± 1.56	89.92 ± 6.04	75.05 ± 3.74	98.24 ± 1.24	$\mathbf{99.95 \pm 0.10}$	99.92 ± 0.13	99.33 ± 0.58	99.71 ± 0.96
8	60.73 ± 1.22	65.16 ± 13.23	73.17 ± 7.54	59.26 ± 6.50	53.15 ± 5.37	82.60 ± 7.36	80.19 ± 4.94	$\mathbf{93.17 \pm 5.05}$
9	97.31 ± 1.48	95.45 ± 3.65	88.30 ± 3.86	97.99 ± 1.72	97.50 ± 1.21	97.74 ± 0.47	91.44 ± 8.49	$\mathbf{99.43 \pm 1.00}$
OA(%)	67.65 ± 1.05	75.52 ± 5.10	71.71 ± 1.39	86.28 ± 1.61	88.27 ± 0.71	91.50 ± 0.68	87.55 ± 2.31	$\mathbf{92.66 \pm 3.16}$
AA(%)	73.89 ± 0.26	76.66 ± 3.95	75.58 ± 1.49	88.62 ± 1.32	88.95 ± 0.68	92.61 ± 0.35	90.84 ± 1.29	$\mathbf{95.19 \pm 1.22}$
$\kappa \times 100$	58.72 ± 1.10	68.75 ± 6.21	63.82 ± 1.64	82.24 ± 2.04	84.71 ± 0.89	88.87 ± 0.86	83.88 ± 2.93	$\mathbf{90.45 \pm 3.99}$

semantic features and has good performance in HSI classification tasks.

7) Memory-augmented spectral-spatial transformer (MASSFormer) [45] represents one of the latest advances in Transformer-based approaches. It leverages the advantages of both CNN and Transformer architectures to propose a novel spectral–spatial feature fusion technique, achieving notable classification performance.

1) Quantitative Results: Tables II–IV present the classification performance of various models on the IP, PU, and WHHH datasets, evaluated using the well-established metrics of OA, AA, κ, and per class accuracy, with the best performance highlighted in bold.

The data in Tables II–IV indicate that our proposed model exhibits the best overall performance across all three datasets. Under the condition of using ten samples per class for training, the performance of the SpectralFormer, HiT, and SS1DSwin models is significantly lower compared to other models. In contrast, the SSSFTT, CASST, LSGA, and MASSFormer models demonstrate high classification accuracy, indicating their strong classification capabilities in small-sample scenarios. Specifically, our model improves the OA by 4.03%, 1.16%, and 5.66% over the second-best results in the IP, PU, and WHHH datasets, respectively.

In analyzing the classification results for detailed categories, we observe that our model outperforms other comparative methods in the majority of categories. Notably, in the PU dataset, the line-type distribution pattern of the Asphalt category significantly differs from other land-cover distribution patterns. The classification accuracy of our proposed algorithm for Asphalt is markedly higher than that of other algorithms (Ours: 89.50%, second-best result CASST: 85.51%), which aligns with the original design intention of our model. Our method outperforms other comparative methods because it introduces two key modules, RSVA and HSVA, which enhance performance in hyperspectral imagery analysis. The RSVA module utilizes random masking techniques to reduce input tokens while preserving important spectral information, allowing the model to efficiently capture spatial information from larger patches without the drawbacks of traditional Transformer-based methods. Meanwhile, the HSVA module progressively reduces the number of tokens between Transformer layers, ensuring crucial spectral features are retained while expanding spatial information extraction. This hierarchical processing enables the model to attend to both local and global information, leading to significant performance gains with larger patch sizes, as demonstrated on the PU and WHHH datasets. Our method maintains high classification accuracy even with larger inputs and shows strong generalization across datasets with varying spatial resolutions. The

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TABLE IV
CLASSIFICATION RESULTS OF EXPERIMENTS ON THE WHHH DATASET (TEN LABELS PER CLASS)

Class	SpectralFormer(2022)	Hit(2022)	SS1DSwin(2023)	SSFTT(2022)	CASST(2022)	LSGA(2023)	MASSFormer(2024)	Ours
1	77.43 ± 0.89	$\mathbf{93.93 \pm 1.16}$	88.89 ± 3.13	91.93 ± 1.04	93.03 ± 0.49	93.88 ± 1.69	92.88 ± 4.20	91.93 ± 2.00
2	52.20 ± 6.45	73.79 ± 6.87	82.07 ± 1.31	81.52 ± 4.43	78.66 ± 2.88	$\mathbf{87.11 \pm 2.51}$	81.03 ± 9.09	81.30 ± 6.26
3	78.44 ± 3.08	78.99 ± 3.86	72.94 ± 1.71	80.72 ± 0.68	82.57 ± 0.45	82.16 ± 2.03	81.26 ± 3.29	$\mathbf{86.12 \pm 1.55}$
4	69.03 ± 7.81	92.16 ± 0.75	84.48 ± 6.84	90.73 ± 1.06	93.12 ± 0.78	90.59 ± 2.21	91.63 ± 3.10	$\mathbf{98.34 \pm 0.40}$
5	67.44 ± 0.62	86.60 ± 2.55	70.61 ± 10.30	92.68 ± 2.32	93.12 ± 1.39	92.28 ± 2.06	87.39 ± 4.29	$\mathbf{98.40 \pm 0.79}$
6	65.68 ± 1.30	83.80 ± 2.79	79.27 ± 4.31	91.84 ± 0.74	91.96 ± 0.76	91.84 ± 0.72	91.38 ± 2.09	$\mathbf{93.14 \pm 0.34}$
7	35.27 ± 1.68	50.72 ± 3.38	52.45 ± 7.26	66.44 ± 2.71	68.16 ± 4.33	68.28 ± 2.75	66.70 ± 8.33	$\mathbf{68.57 \pm 3.46}$
8	23.29 ± 2.17	33.90 ± 15.23	35.30 ± 2.90	65.44 ± 3.07	69.02 ± 2.02	84.07 ± 1.56	75.30 ± 4.25	$\mathbf{93.17 \pm 5.21}$
9	88.74 ± 1.70	82.86 ± 8.08	88.12 ± 7.64	93.85 ± 0.87	94.79 ± 0.08	94.13 ± 0.29	$\mathbf{96.11 \pm 3.28}$	94.43 ± 1.39
10	37.48 ± 10.71	33.34 ± 9.86	53.81 ± 10.71	70.90 ± 7.99	78.94 ± 1.00	75.44 ± 3.64	73.77 ± 13.50	$\mathbf{83.29 \pm 2.58}$
11	28.04 ± 2.21	23.10 ± 5.77	46.22 ± 1.29	61.42 ± 3.28	66.52 ± 1.32	70.13 ± 5.55	55.99 ± 13.81	$\mathbf{76.72 \pm 9.88}$
12	56.88 ± 5.24	60.77 ± 5.87	54.16 ± 3.00	58.37 ± 5.38	70.54 ± 1.82	64.43 ± 3.52	59.08 ± 9.36	$\mathbf{80.10 \pm 3.93}$
13	34.53 ± 4.64	47.01 ± 5.83	48.43 ± 4.08	48.20 ± 7.27	43.28 ± 3.87	61.22 ± 4.73	60.73 ± 10.69	$\mathbf{75.67 \pm 5.99}$
14	57.56 ± 2.18	60.74 ± 5.73	70.60 ± 9.26	92.56 ± 2.38	$\mathbf{96.20 \pm 0.63}$	68.95 ± 1.08	84.61 ± 11.44	91.37 ± 1.85
15	74.54 ± 3.70	92.24 ± 2.56	86.36 ± 5.32	98.13 ± 1.11	98.19 ± 0.61	96.83 ± 1.07	98.39 ± 1.28	$\mathbf{99.51 \pm 0.61}$
16	57.76 ± 9.71	27.08 ± 9.00	68.34 ± 5.34	90.44 ± 1.62	84.22 ± 1.64	$\mathbf{97.23 \pm 1.89}$	83.36 ± 17.25	91.84 ± 1.24
17	71.41 ± 3.52	74.27 ± 10.87	74.02 ± 3.22	96.87 ± 0.85	96.54 ± 0.60	95.03 ± 0.64	83.03 ± 11.00	$\mathbf{99.19 \pm 0.53}$
18	33.93 ± 18.76	63.04 ± 10.48	70.23 ± 2.80	$\mathbf{99.40 \pm 0.34}$	99.19 ± 0.30	98.30 ± 0.50	96.39 ± 3.56	99.03 ± 0.60
19	67.34 ± 5.83	57.44 ± 7.63	78.81 ± 2.47	89.43 ± 1.84	90.67 ± 2.54	85.73 ± 4.47	83.49 ± 9.31	$\mathbf{90.83 \pm 1.80}$
20	77.72 ± 4.09	74.39 ± 1.74	80.84 ± 11.67	83.14 ± 5.55	90.34 ± 1.32	$\mathbf{96.67 \pm 0.53}$	90.08 ± 7.24	92.65 ± 2.08
21	49.88 ± 4.17	63.57 ± 10.32	87.08 ± 2.01	91.53 ± 2.88	93.12 ± 3.38	94.83 ± 1.61	94.05 ± 4.85	$\mathbf{99.58 \pm 0.63}$
22	53.72 ± 3.84	86.23 ± 4.55	78.00 ± 4.42	86.53 ± 5.03	88.32 ± 0.84	95.23 ± 1.80	90.73 ± 5.67	$\mathbf{99.47 \pm 1.08}$
OA(%)	61.93 ± 4.44	76.31 ± 1.34	74.88 ± 4.45	83.93 ± 0.36	85.71 ± 0.43	85.47 ± 1.64	84.66 ± 2.58	$\mathbf{91.36 \pm 0.57}$
AA(%)	57.20 ± 1.29	65.45 ± 1.75	70.50 ± 2.80	82.82 ± 0.68	84.57 ± 0.39	85.65 ± 1.07	82.61 ± 1.81	$\mathbf{90.21 \pm 0.55}$
$\kappa \times 100$	55.18 ± 4.43	70.65 ± 1.66	69.45 ± 5.01	80.10 ± 0.42	82.24 ± 0.52	82.02 ± 1.95	80.99 ± 3.03	$\mathbf{89.14 \pm 0.71}$

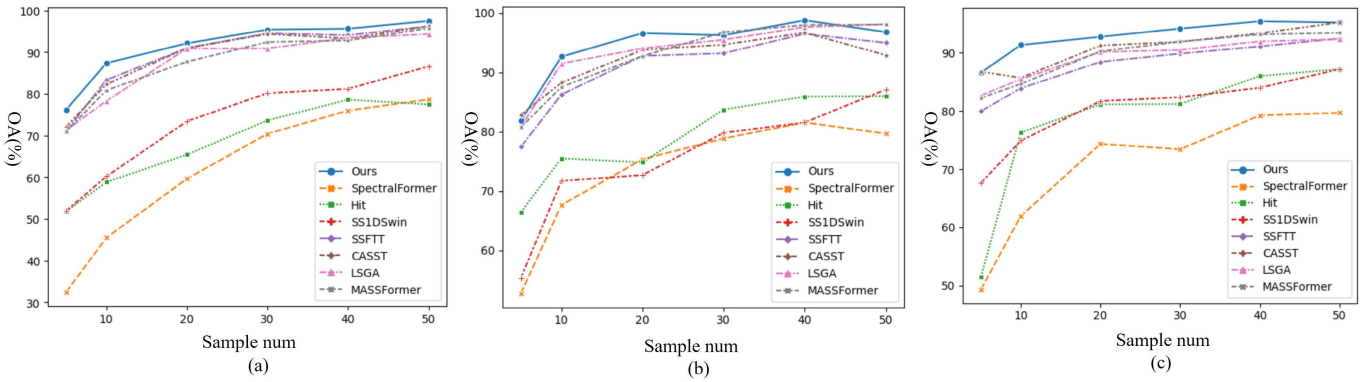


Fig. 10. OA of different methods on the IP, PU, and WHHH datasets with varying training sample sizes. (a) IP. (b) PU. (c) WHHH.

combined effectiveness of RSVA and HSVA, along with the management of larger patch sizes, underpins our method's superior performance compared to existing approaches.

We also validated the performance of our model with varying numbers of training samples, as shown in Fig. 10. We adjusted the sample size for training each dataset to evaluate the classification performance of different methods. Fig. 10 demonstrates that as the sample size increases, the OA of all methods improves. Notably, our model consistently exhibits outstanding performance across different training sample sizes in the IP, PU, and WHHH datasets, often achieving the best classification results. This experiment further validates the superiority of our model.

2) Visual Evaluation: Figs. 11–13 present the final classification results for the IP, PU, and WHHH datasets, respectively. Our model demonstrates the least noise and the best classification results. Overall, the SpectralFormer, HiT, and SS1DSwin models exhibit more misclassifications, particularly within clusters of similar land-cover types. For instance, in Fig. 11, the images produced by these three models appear most fragmented. In contrast, the SSFTT, CASST, and LSGA methods significantly reduce noise in the classification results. Building on these improvements, our model further enhances classification performance, particularly in scenes with high land-cover density within a unit area (e.g., the upper-left region of the IP dataset, the area around the bricks in the PU dataset, and the lower-right region of the WHHH dataset). In these scenarios, our model shows superior OA compared to other methods, demonstrating exceptional precision in challenging environments.

C. Ablation Study

To further validate the effectiveness of the RSVA and HSVA block in the SVAFormer model, we conducted an extended ablation study. We performed experiments on three standard hyperspectral datasets: IP, PU, and WHHH datasets, the same as in the previous experiment. For each dataset, we measured performance using OA, AA, and κ. All models were trained with the same hyperparameters, and the results were averaged in five runs to ensure statistical significance. To explore

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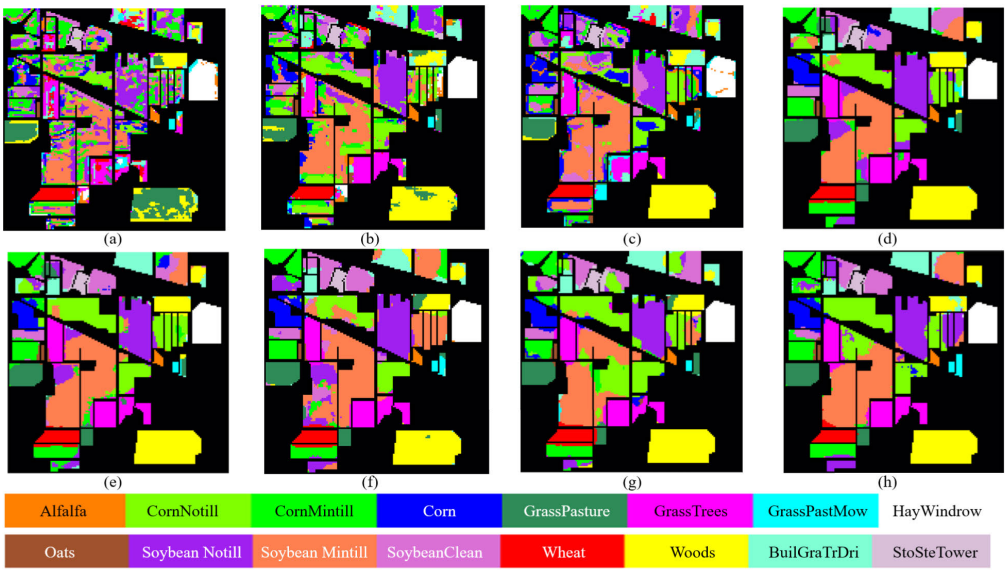


Fig. 11. Visual classification results on the IP dataset with compared methods. (a) SpectralFormer. (b) HiT. (c) SS1DSwin. (d) SSFTT. (e) CASST. (f) LSGA. (g) MASSFormer. (h) Ours.

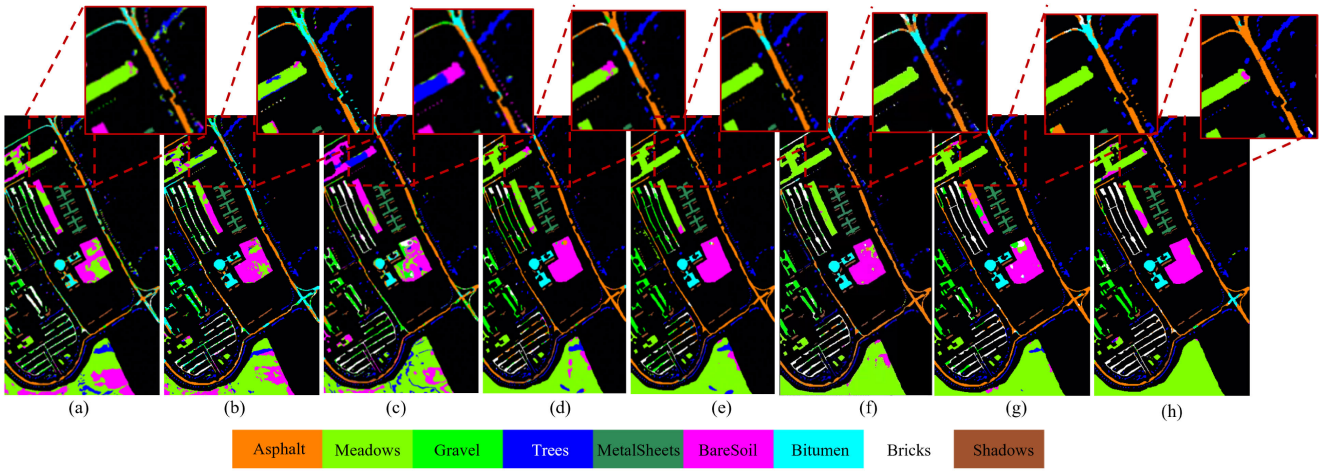


Fig. 12. Visual classification results on the PU dataset with compared methods. (a) SpectralFormer. (b) HiT. (c) SS1DSwin. (d) SSFTT. (e) CASST. (f) LSGA. (g) MASSFormer. (h) Ours.

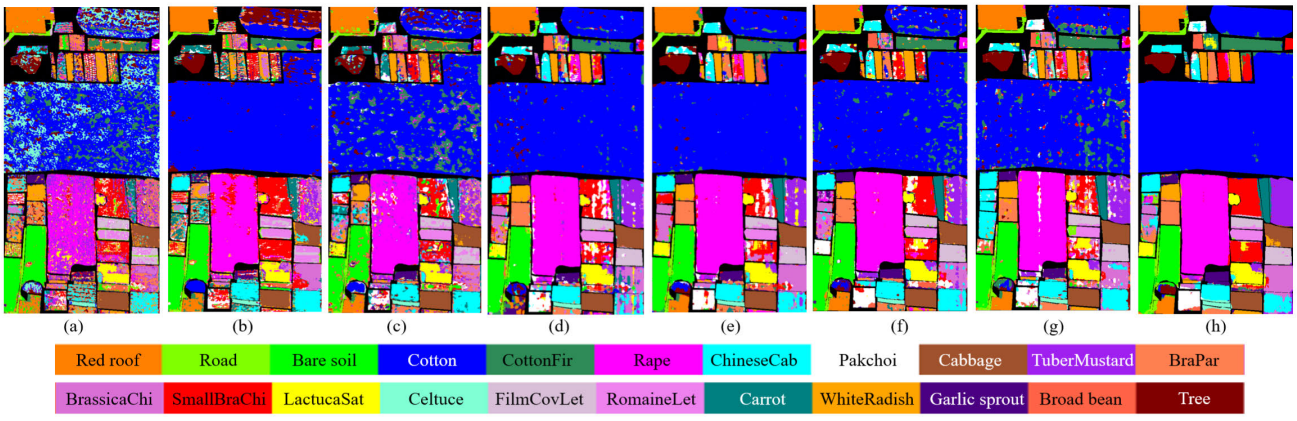


Fig. 13. Visual classification results on the WHHH dataset with compared methods. (a) SpectralFormer. (b) HiT. (c) SS1DSwin. (d) SSFTT. (e) CASST. (f) LSGA. (g) MASSFormer. (h) Ours.

the effects of the two modules in the model, we designed experiments under four different scenarios.

1) Remove the RSVA module and the HSVA module.
2) Only remove the RSVA module.

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TABLE V
CLASSIFICATION PERFORMANCE OF THE PROPOSED CASST METHOD WITH DIFFERENT ABLATION STRATEGIES ON THE THREE HSI DATASETS

Index	RSVA	HSVA	Indian Pines			Pavia University			WHU-HI-HONGHU		
			OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$	OA(%)	AA(%)	$\kappa \times 100$
1	×	×	85.53 ± 2.40	91.99 ± 0.75	83.69 ± 2.61	90.01 ± 3.24	93.88 ± 1.12	87.10 ± 4.02	89.46 ± 0.99	88.19 ± 1.01	86.78 ± 1.22
2	×	✓	86.52 ± 2.40	92.73 ± 0.77	84.80 ± 2.63	92.23 ± 4.55	95.03 ± 1.49	89.92 ± 5.75	90.68 ± 0.60	89.56 ± 0.59	88.29 ± 0.74
3	✓	×	86.01 ± 2.16	92.56 ± 0.77	84.23 ± 2.37	90.77 ± 2.00	91.84 ± 1.94	87.87 ± 2.64	89.41 ± 0.95	87.92 ± 0.80	86.71 ± 1.15
4	✓	✓	87.33 ± 2.30	93.26 ± 0.76	85.70 ± 2.54	92.66 ± 3.16	95.19 ± 1.22	90.45 ± 3.99	91.36 ± 0.57	90.21 ± 0.55	89.14 ± 0.71

· RSVA: Random Spectral View Attention Mechanism. HSVA: Hierarchical Spectral View Attention Mechanism.

3) Only remove the HSVA module.
4) Our proposed SVAFormer model.

The detailed experimental results are shown in Table V. By comparing Experiment 4 with Experiment 1, we can observe that our algorithm improves by 1.80%, 2.65%, and 1.90% on the IP, PU, and WHHH datasets, respectively, compared to the backbone ViT model, validating the effectiveness of our proposed improvements. Additionally, to further verify the individual impacts of RSVA and HSVA, we conducted Experiments 2 and 3. A comparison of Experiment 4 with Experiment 2, where the RSVA structure is removed, shows that the OA of our model decreases by 0.81%, 0.43%, and 0.68% on the IP, PU, and WHHH datasets, respectively. Similarly, a comparison of Experiment 4 with Experiment 3, where the HSVA structure is removed, reveals that the OA of our model decreases by 1.32%, 1.89%, and 1.95% on the IP, PU, and WHHH datasets, respectively. It can be observed that the OA loss resulting from the removal of the HSVA structure is significantly higher than that from the removal of the RSVA structure.

IV. CONCLUSION

Traditional HSI classification methods based on Transformers have faced challenges in handling the widely varying scales and diverse spatial distribution patterns commonly present in HSIs. In particular, most Transformer frameworks that divide pixels into tokens exhibit a performance trend where classification accuracy initially improves but subsequently declines as the patch size increases. This phenomenon is primarily due to the increase in sequence length, which hampers the attention mechanism from effectively focusing on the spectral features of the central pixel and the spatial features of neighboring pixels. This issue becomes more pronounced when dealing with land covers of different scales and distribution patterns. To address these challenges, we propose the SVAFormer framework, which includes two crucial components: RSVA and HSVA. RSVA employs a random masking technique to provide spectral tokens with spatial neighborhood information from various perspectives, enabling the model to handle diverse object distribution patterns. HSVA introduces a spectral token-aware pooling layer between adjacent Transformer blocks, allowing the model to expand the spatial scale represented by each token while maintaining the central position of the spectral token. This enables the spectral token to capture spatial neighborhood information across different levels and scales. Finally, we validated the effectiveness of our framework on three public datasets.

Although the SVAFormer framework shows strong performance in addressing multiscale and multimorphology challenges in HSI classification, it still has limitations. The RSVA module's random masking is sensitive to different datasets, which may affect its ability to capture spatial information. The HSVA module, while expanding spatial scale, may lose information when handling larger objects. Moreover, the high computational cost of Transformer models limits their use in resource-constrained scenarios.

In future work, we will explore more efficient random masking strategies to improve adaptability, optimize the HSVA module for handling larger-scale objects, and investigate lightweight Transformer techniques to reduce computational costs. We also plan to explore broader sequence processing structures, such as Mamba, and advanced methods like self-supervised contrastive learning to enhance the model's generalization. These improvements aim to make SVAFormer more effective in complex hyperspectral data environments.

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